

torch.map#

<https://pytorch.org/docs/stable/generated/exportdb/torch.map.html>

Description:

Tags: torch.dynamic-shape, torch.map Support Level: SUPPORTED

Original source code:

Code Examples

```

# mypy: allow-untyped-defs
import torch

from functorch.experimental.control_flow import map

class DynamicShapeMap(torch.nn.Module):
    """
    functorch map() maps a function over the first tensor
    dimension.
    """

    def forward(self, xs, y):
        def body(x, y):
            return x + y

        return map(body, xs, y)

example_args = (torch.randn(3, 2), torch.randn(2))
tags = {"torch.dynamic-shape", "torch.map"}
model = DynamicShapeMap()

torch.export.export(model, example_args)

```

ExportedProgram:

```
class GraphModule(torch.nn.Module):
    def forward(self, xs: "f32[3, 2]", y: "f32[2]"):
        body_graph_0 = self.body_graph_0
        map_impl =
torch.ops.higher_order.map_impl(body_graph_0, [xs], [y]);
body_graph_0 = xs = y = None
    getitem: "f32[3, 2]" = map_impl[0]; map_impl = None
    return (getitem,)

class body_graph_0(torch.nn.Module):
    def forward(self, xs: "f32[2]", y: "f32[2]"):
        add: "f32[2]" =
torch.ops.aten.add.Tensor(xs, y); xs = y = None
    return (add,)
```

Graph signature:

inputs

xs: USER_INPUT

y: USER_INPUT

outputs

getitem: USER_OUTPUT

Range constraints: {}

Notes & Warnings

NOTE

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Detailed Documentation

`dynamic_shape_map#`

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